

Cytiva Xcellerex X-platform 50 L - downflow configuration

cytiva-x-platform-50

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	10 L	Source-supported
Maximum working volume	50 L	Source-supported
Nominal vessel / bag size	50 L	Source-supported
Total vessel volume loaded into the form	50 L	Derived / estimated
Initial working volume loaded into the form	10 L	Derived / estimated
Product-contact material	ReadyKleer single-use film	Source-supported
Height : diameter	2	App assumption
Internal / chamber diameter	0.3169 m	Derived / estimated
Liquid height at maximum working volume	0.6338 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	App assumption
Impeller count	1	Source-supported
Impeller diameter	0.1046 m	Derived / estimated
Power number	0.4	App assumption
Initial agitation	100 rpm	App assumption
Maximum agitation	285 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	Yes	Source-supported
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 10 L

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 5; System specifications, X-50

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Direct working range and downflow agitation ceiling. X-200 upflow ceiling is instead 215 rpm.

Local original: [Cytiva Xcellerex X-platform datasheet - PDF page 5](#)

[Manufacturer publication](#)

Maximum working volume - 50 L

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 5; System specifications, X-50

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: Direct working range and downflow agitation ceiling. X-200 upflow ceiling is instead 215 rpm.

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Nominal vessel / bag size - 50 L

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 5; System specifications, X-50

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: Direct working range and downflow agitation ceiling. X-200 upflow ceiling is instead 215 rpm.

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Total vessel volume loaded into the form - 50 L

Derived / estimated | Form-population rule

Rule: total volume = $\max(\text{nominal vessel volume}, \text{maximum working volume})$

Selection / calculation: $\max(50 \text{ L}, 50 \text{ L}) = 50 \text{ L}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 10 L

Derived / estimated | Form-population rule

Rule: initial working volume = $\min(\text{minimum working volume}, \text{maximum working volume})$

Selection / calculation: $\min(10 \text{ L}, 50 \text{ L}) = 10 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - ReadyKleer single-use film

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 3; Bioreactor bags

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: ReadyKleer single-use film.

Limitations: Film brand documented; specific contact-layer polymer not established here.

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[Manufacturer publication](#)

Height : diameter - 2

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].heightDiameterRatio = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.3169 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.05 \text{ m}^3) / (\pi \times 2)} = 0.3169203 \text{ m}$; the form displays 0.3169 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.6338 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.3169203 \text{ m} \times 2 = 0.6338406 \text{ m}$; the form displays 0.6338 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Pitched-blade turbine

App assumption | Cytiva Xcellerex X-platform datasheet - PDF page 5; Impeller type

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Manufacturer designation is 6B-R50 (PDF p. 5). Pitched-blade is an unvalidated simulation approximation; generic power number is not manufacturer-confirmed.

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[Manufacturer publication](#)

Impeller count - 1

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 2; Bioreactor vessel

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.

Limitations: Single bottom-centred drive and impeller; baffles described.

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[Manufacturer publication](#)

Impeller diameter - 0.1046 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.3169203 \text{ m}, 0.001 \text{ m}) = 0.1045837 \text{ m}$; the form displays 0.1046 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 0.4

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].powerNumber = 0.4

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 100 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseRpm = 100`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 285 rpm

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 5; System specifications, X-50

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 285 rpm.

Limitations: Direct working range and downflow agitation ceiling. X-200 upflow ceiling is instead 215 rpm.

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Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Source-supported | Cytiva Xcellerex X-platform datasheet - PDF page 2; Bioreactor vessel

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Yes.

Limitations: Single bottom-centred drive and impeller; baffles described.

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[Manufacturer publication](#)

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Cytiva Xcellerex X-platform datasheet

cytiva-xcellerex-x-platform-datasheet-5b760c05911a.pdf

SHA-256: 5b760c05911a7416765306cb39ea772cec608ad167923b451ead47047b5de9a1